

# Bayesian Graphical Modeling

## Network Comparisons

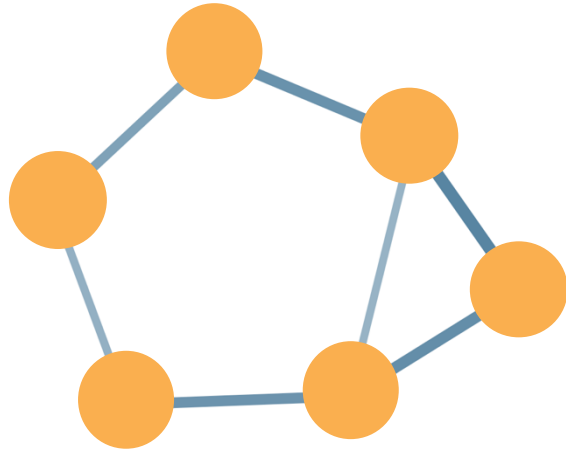
*PsyNets Winterschool 2026*

[m.marsman@uva.nl](mailto:m.marsman@uva.nl)

[bayesiangraphicalmodeling.com](http://bayesiangraphicalmodeling.com)

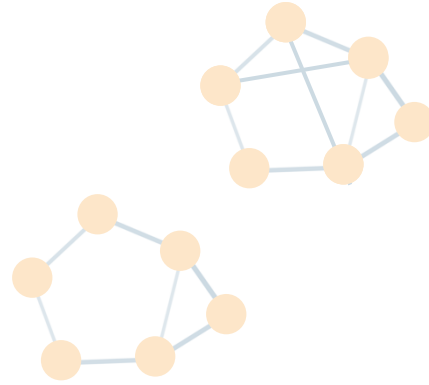


# Setup Today



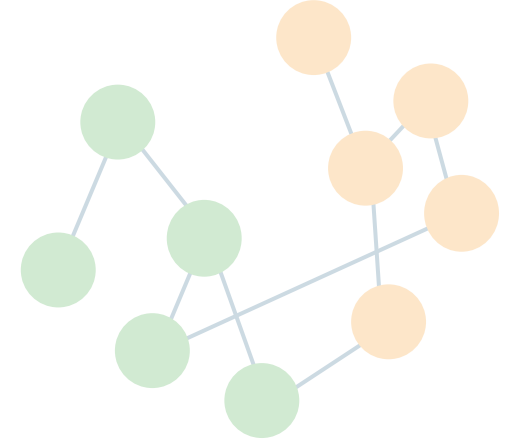
Estimating & Testing

Morning Session



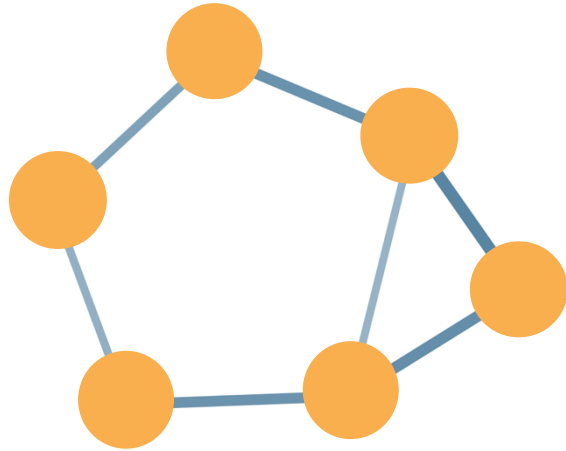
Network Comparison

Afternoon Session



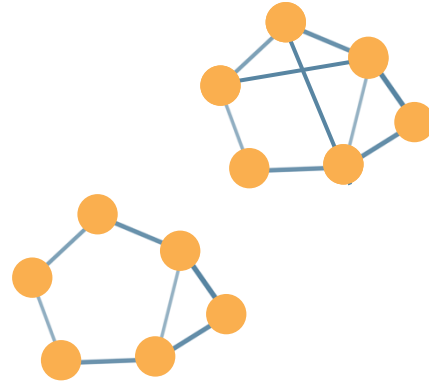
Clustering

# Setup Today

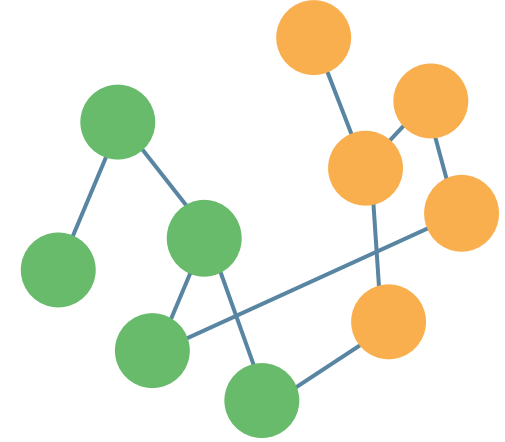


Estimating & Testing

Morning Session



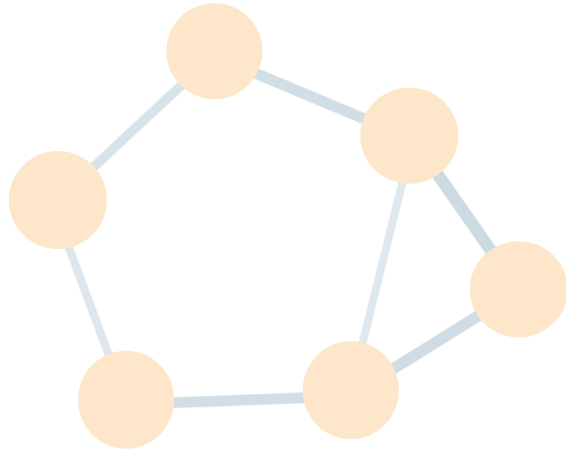
Network Comparison



Clustering

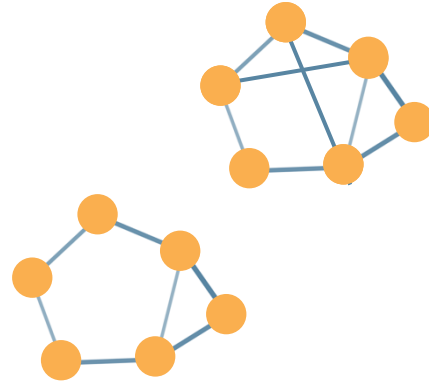
Afternoon Session

# Setup Today



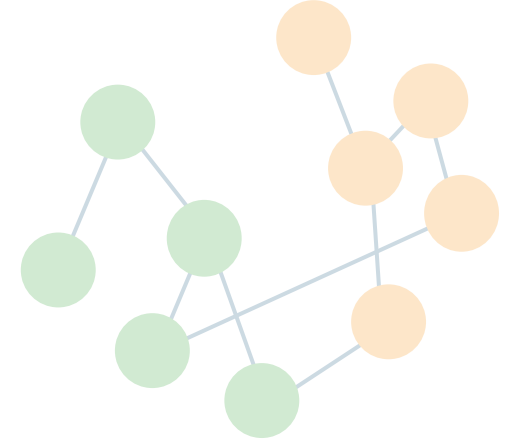
Estimating & Testing

Morning Session



Network Comparison

Afternoon Session



Clustering

# Session setup



Part I: Theory



Part II: Practical



Part III: Project

# Session setup



## Part I: Theory

# Why Compare Networks?

# Why Compare Networks?

## 1. Assess network variance or invariance

- Do network structures (response behavior) differ between groups?
- Which relations are stable (invariant) across groups?
- Where do key differences emerge?

This is a **psychometric approach**, very similar to questions about *differential item functioning* or *measurement invariance* in IRT/SEM frameworks.



# Why Compare Networks?

## 1. Assess network variance or invariance

## 2. Reproducibility

- Do network results reproduce in different samples?
- Do network results generalize to different populations?

A central motivation for developing Bayesian approaches to group comparisons is the concern about *reproducibility* in network psychometrics.

This is a **methodological approach**, aiming to distinguish signal from noise.

# Why Compare Networks?

1. Assess network variance or invariance

2. Reproducibility

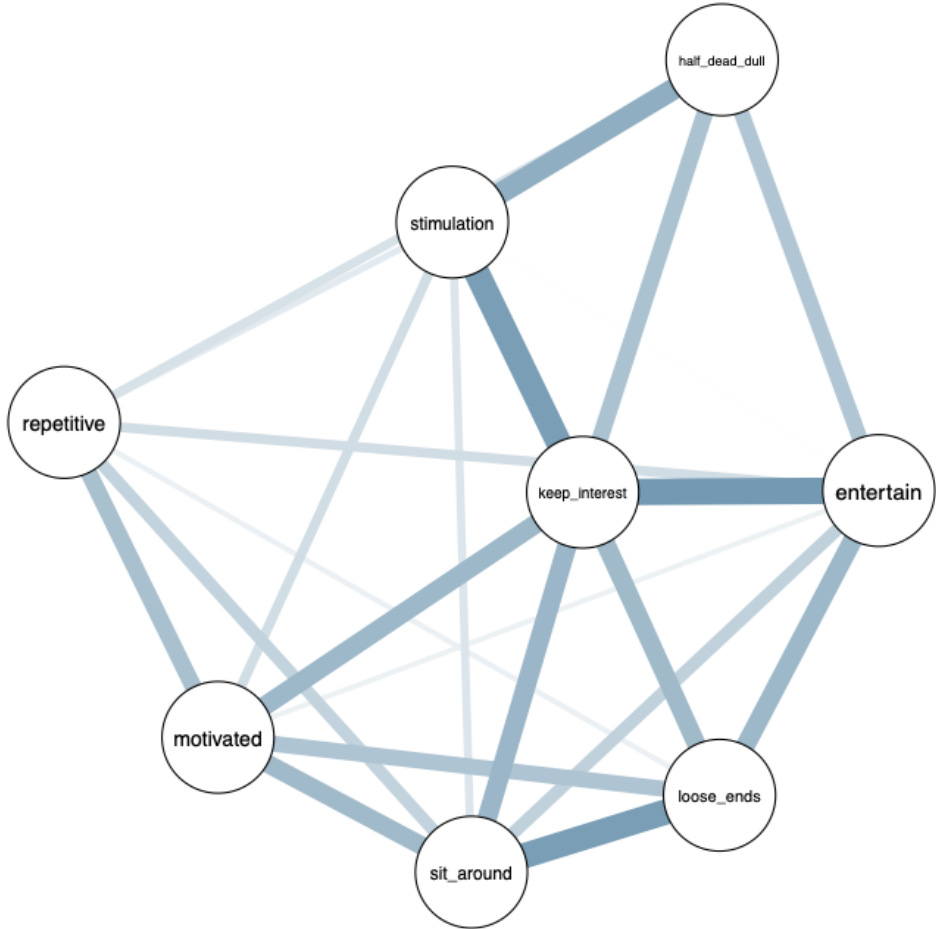
3. Assess theoretical predictions

- Do empirical networks match theoretical expectations?

*For example, expected subgroup differences (e.g., involvement)*

Network models are not only descriptive; they can also be used to formalize and test theories. This offers a **substantive perspective** to network comparison.

# Boredom Proneness Network



Nodes are short boredom proneness scale items (7-point Likert-scale).

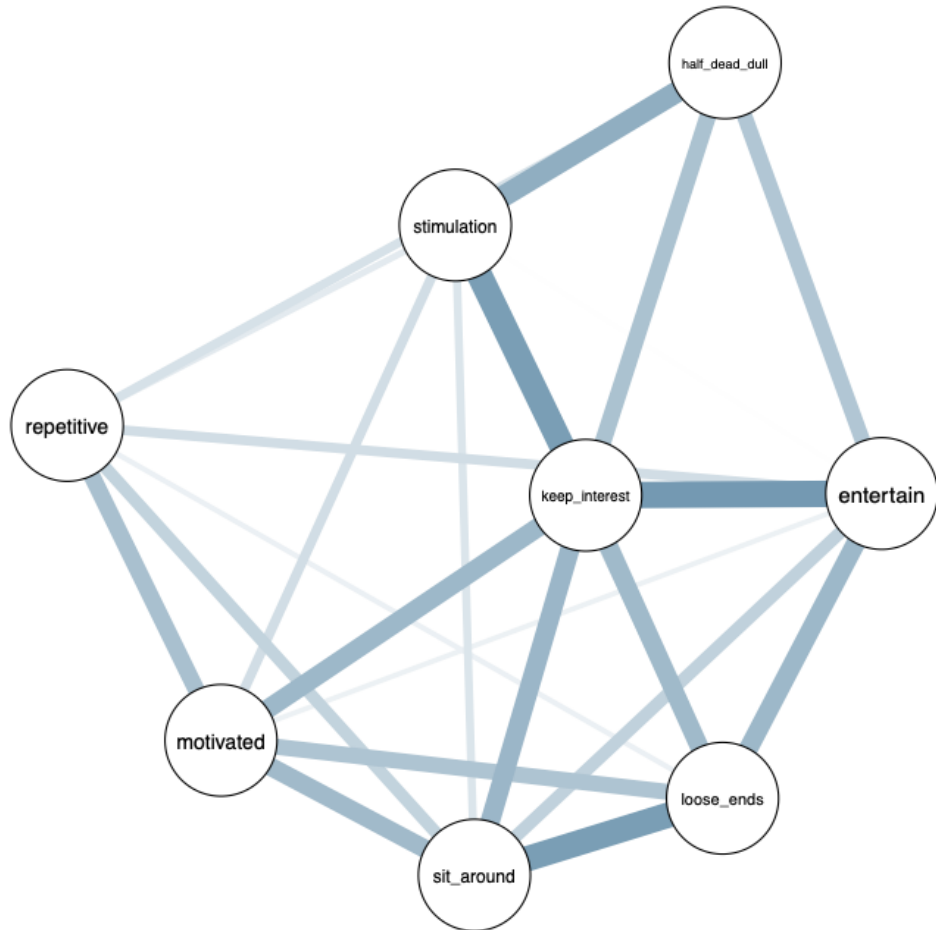
Edges are partial associations.

Goal is to determine conditional dependence relationships.

```
library(bgms)
?Boredom
```

Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, 39(6), 390-399.

# Boredom Proneness Network



Nodes are short boredom proneness scale items (7-point Likert-scale).

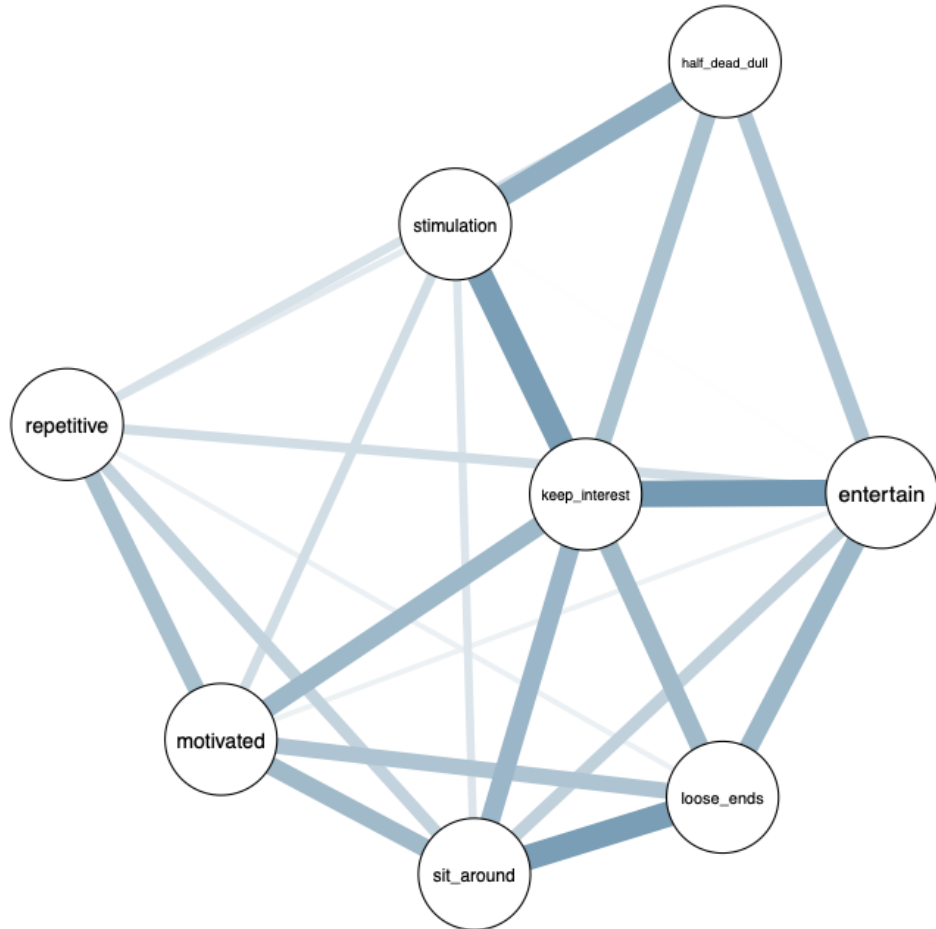
Edges are partial associations.

Goal is to determine conditional dependence relationships.

**library(bgms)**  
?Boredom

Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, 39(6), 390-399.

# Boredom Proneness Network



Nodes are short boredom proneness scale items (7-point Likert-scale).

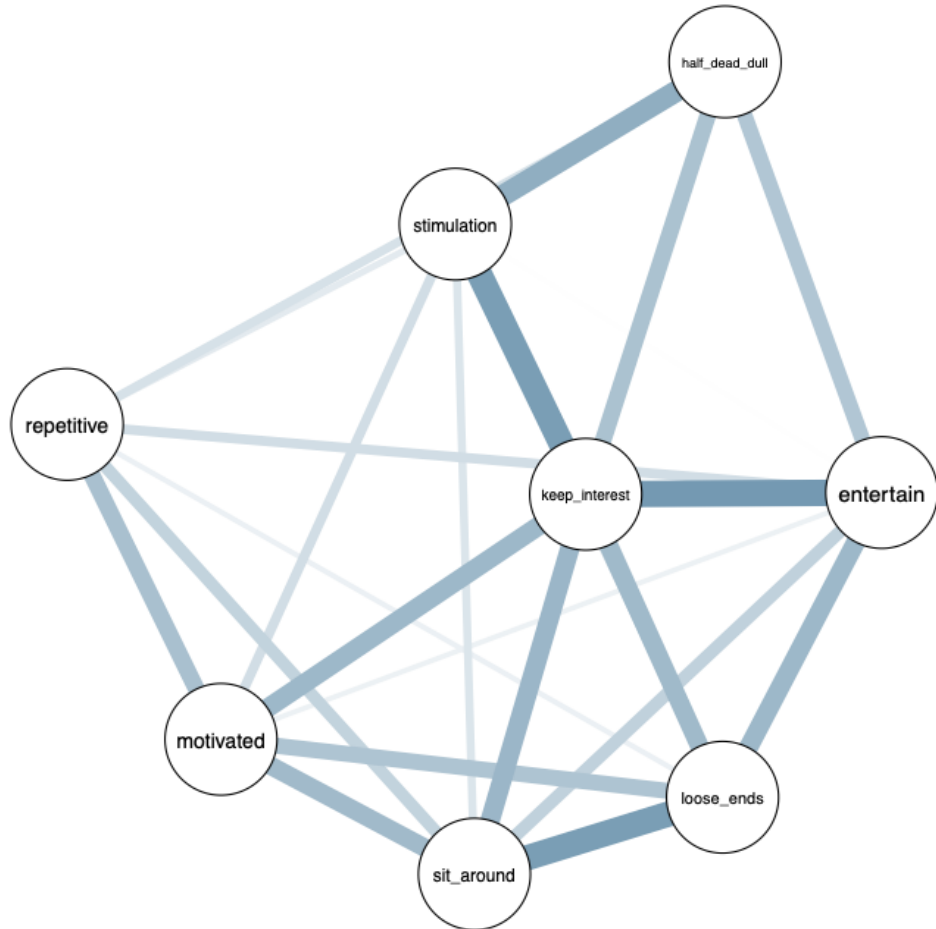
## Edges are partial associations.

Goal is to determine conditional dependence relationships.

```
library(bgms)
?Boredom
```

Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, 39(6), 390-399.

# Boredom Proneness Network



Nodes are short boredom proneness scale items (7-point Likert-scale).

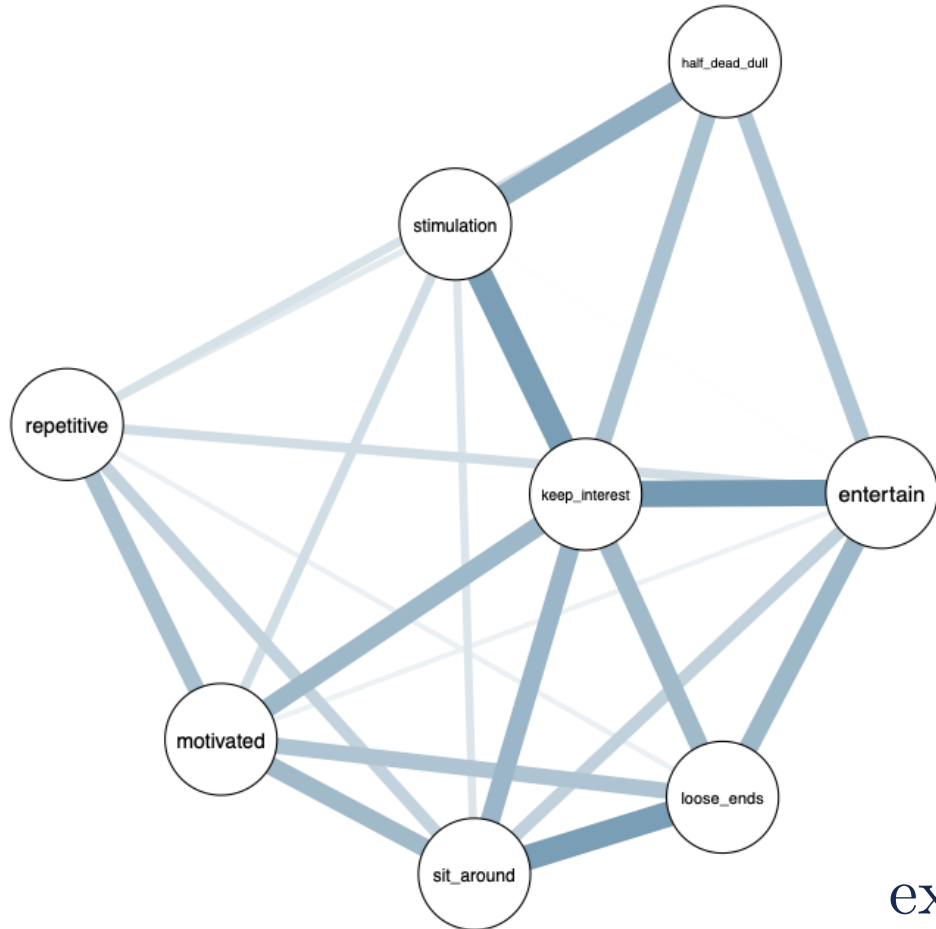
Edges are partial associations.

Goal is to determine conditional dependence relationships.

**library(bgms)**  
**?Boredom**

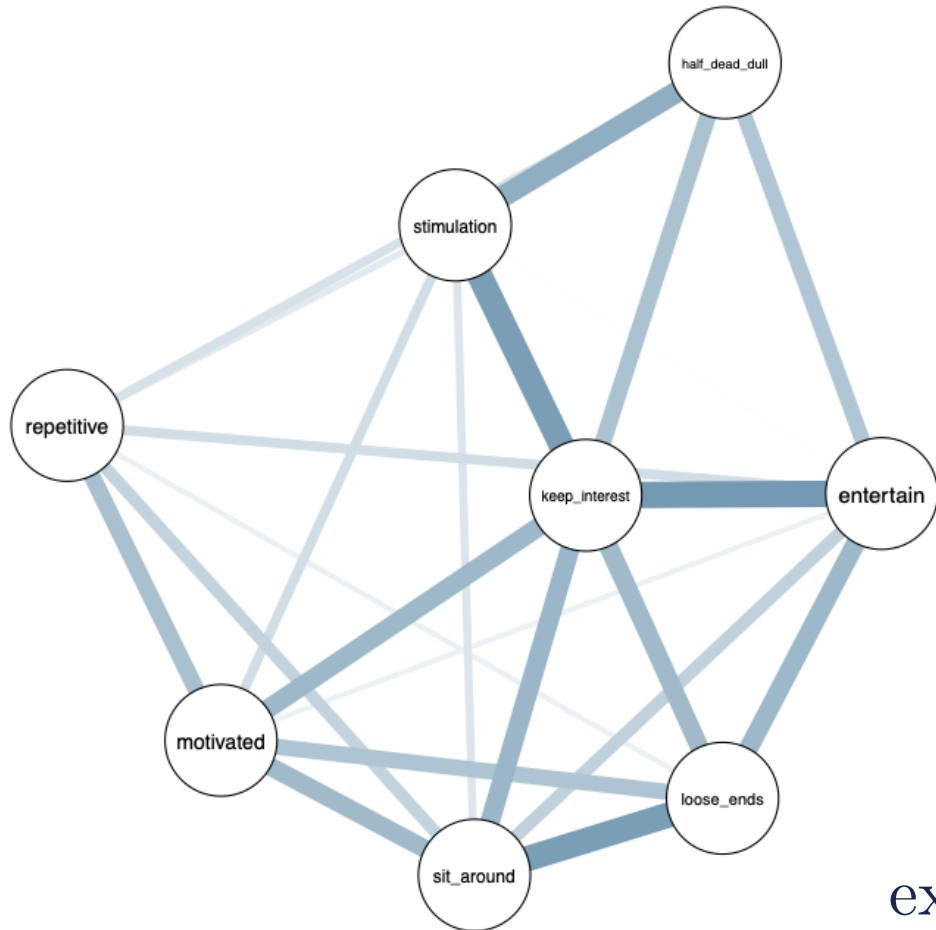
Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, 39(6), 390-399.

# The Ordinal Markov Random Field



$$\frac{\exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}{\sum_{\mathbf{x}} \exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}$$

# The Ordinal Markov Random Field

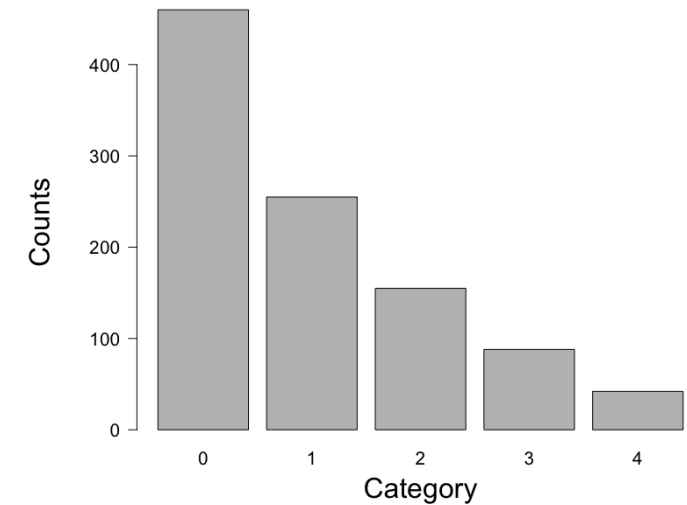
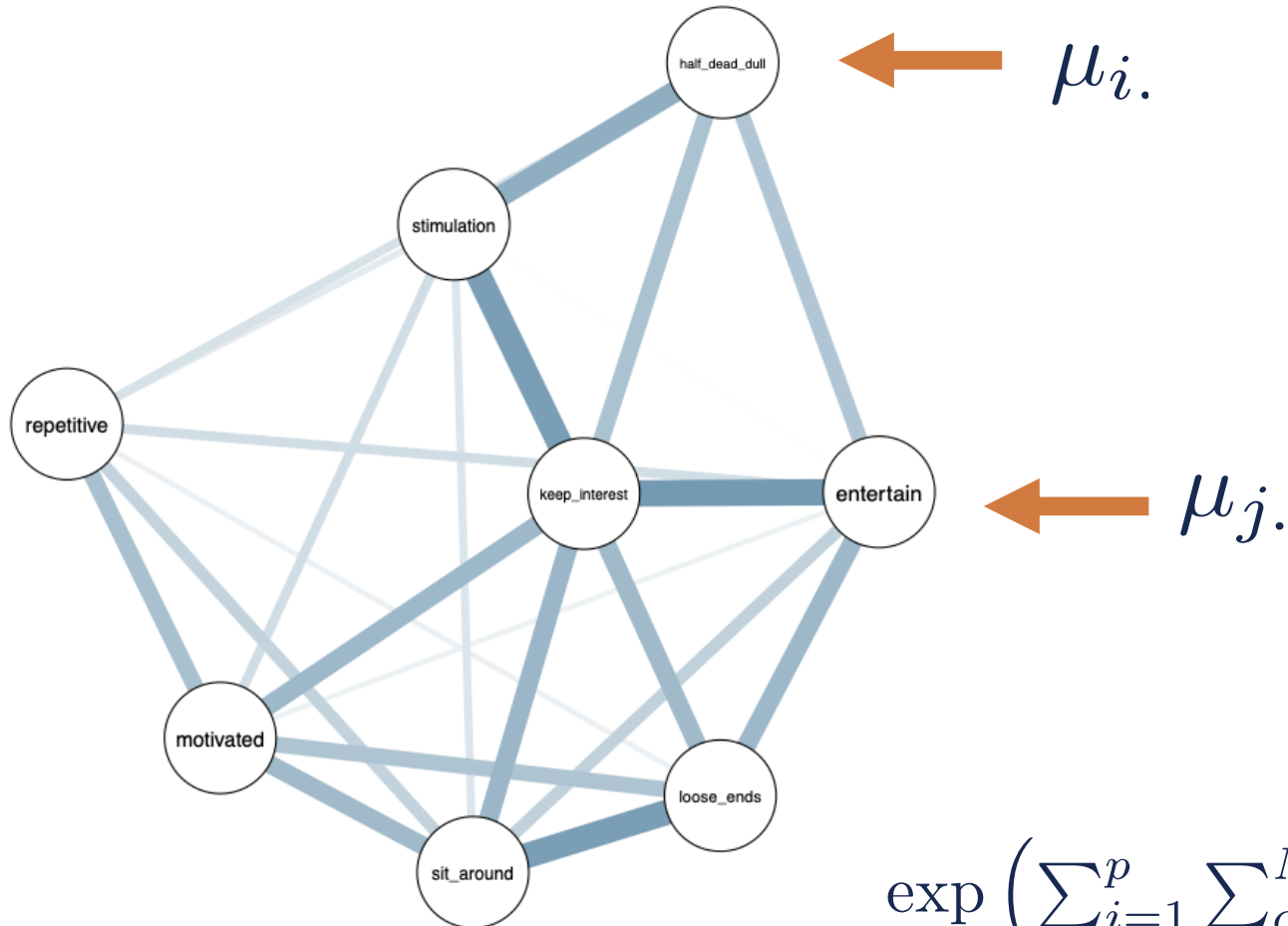


Nodes are binary or ordinal variables.

$$\frac{\exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}{\sum_{\mathbf{x}} \exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}$$

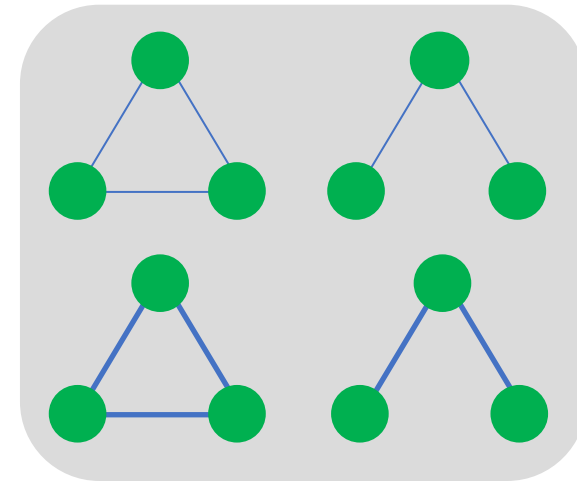
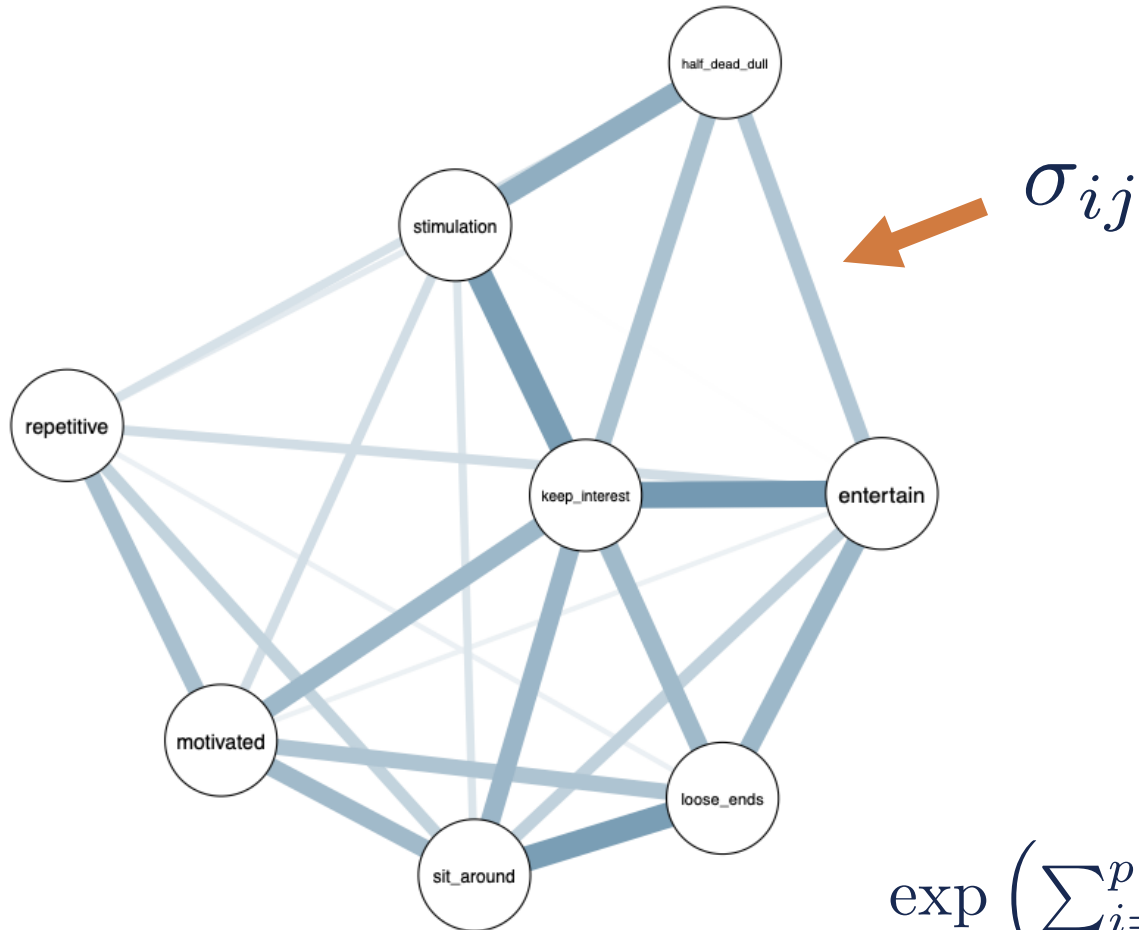


# The Ordinal Markov Random Field



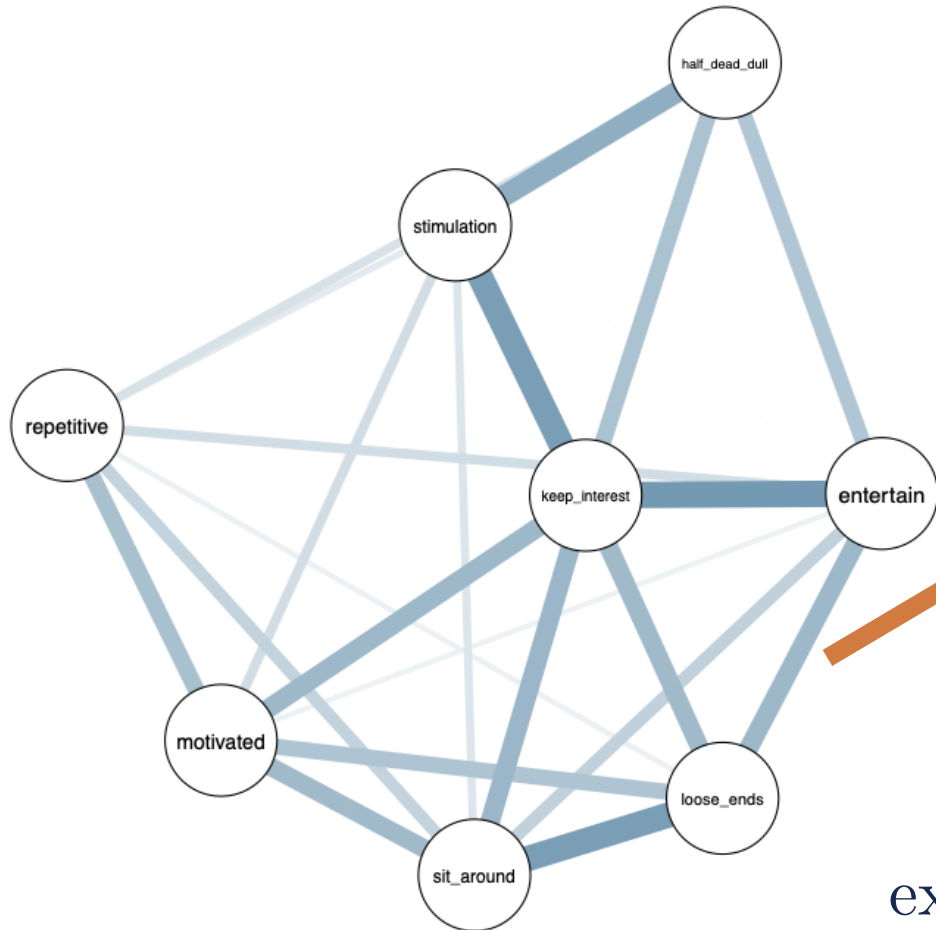
$$\frac{\exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}{\sum_{\mathbf{x}} \exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}$$

# The Ordinal Markov Random Field



$$\frac{\exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}{\sum_{\mathbf{x}} \exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}$$

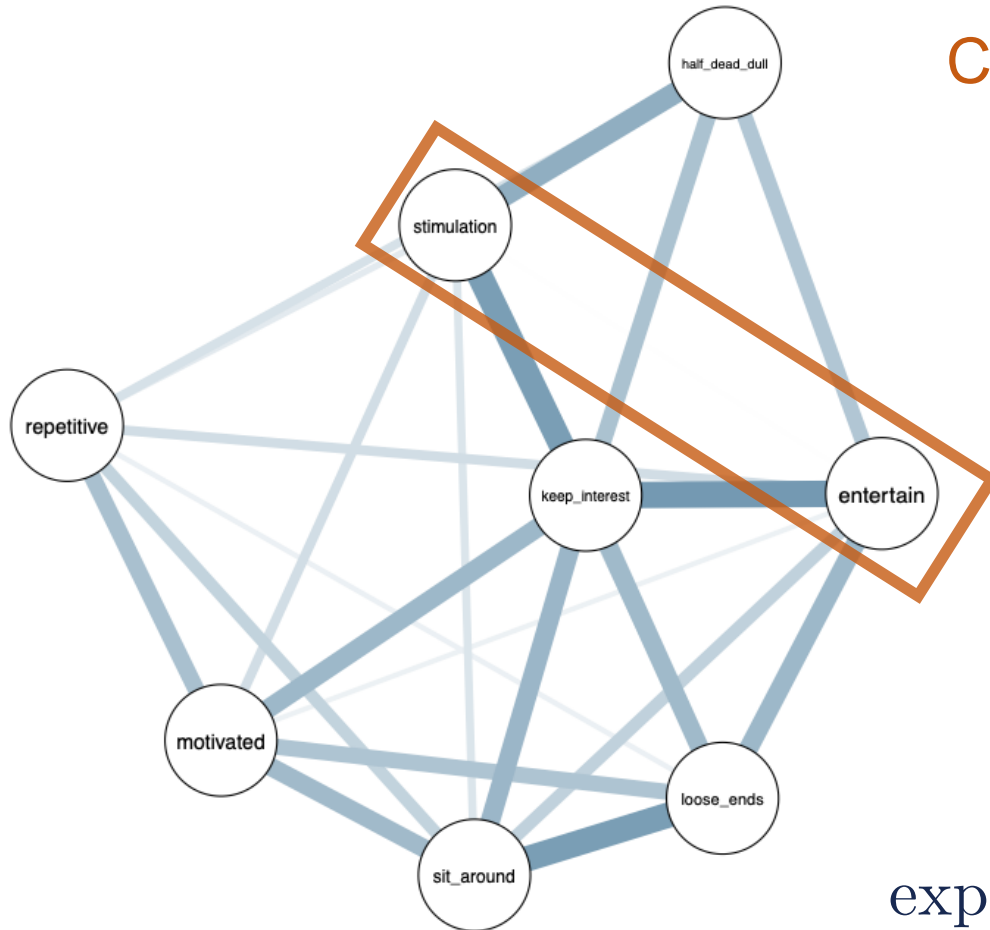
# The Ordinal Markov Random Field



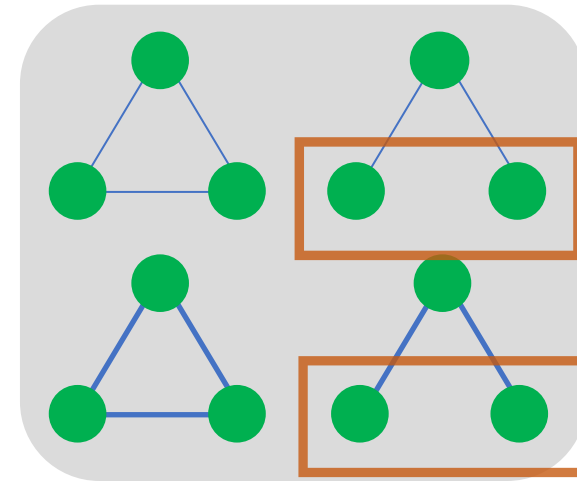
| SPECIFIC OVERVIEW |                        |          |           |             |              |              |
|-------------------|------------------------|----------|-----------|-------------|--------------|--------------|
|                   | Relation               | Estimate | Posterior | Incl. Prob. | Inclusion BF | Category     |
|                   | entertain-loose_ends   | 0.116    |           | 1.000       | Inf          | included     |
|                   | repetitive-loose_ends  | 0.016    |           | 0.353       | 0.546        | inconclusive |
|                   | stimulation-loose_ends | 0.079    |           | 1.000       | Inf          | included     |

$$\frac{\exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}{\sum_{\mathbf{x}} \exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}$$

# The Ordinal Markov Random Field

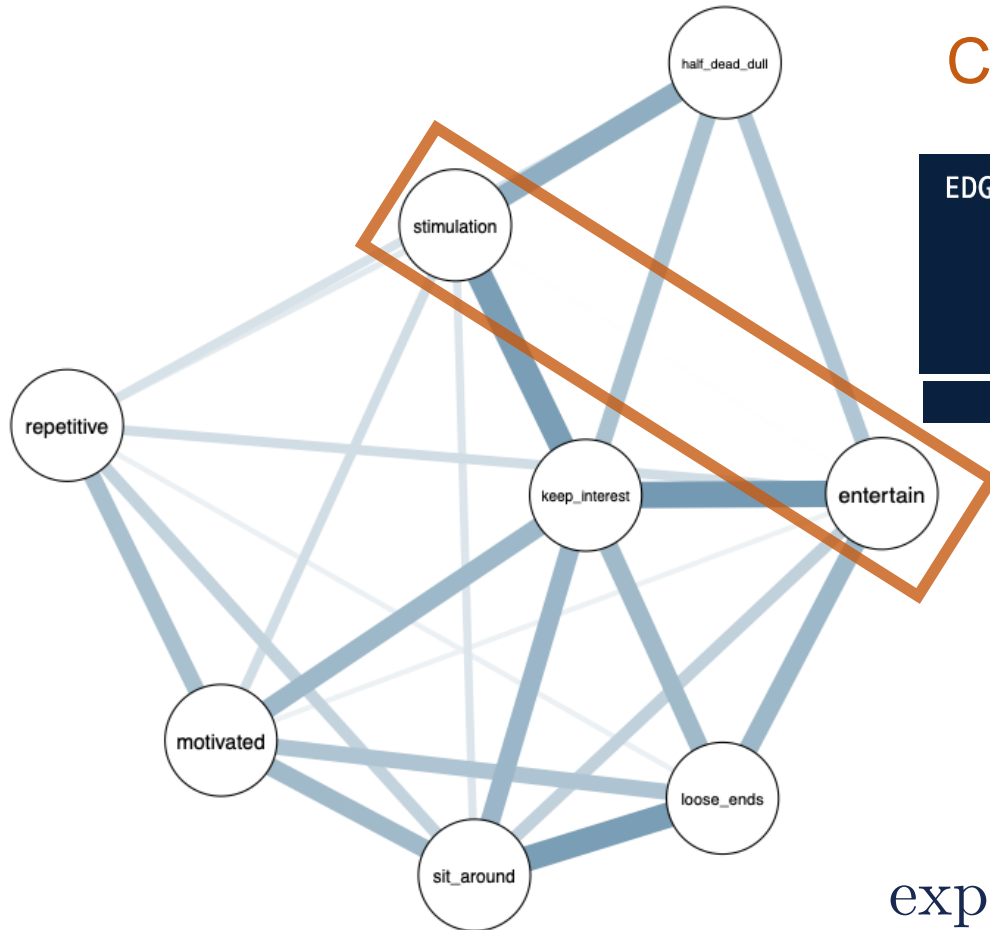


Conditional independence?



$$\frac{\exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}{\sum_{\mathbf{x}} \exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}$$

# The Ordinal Markov Random Field

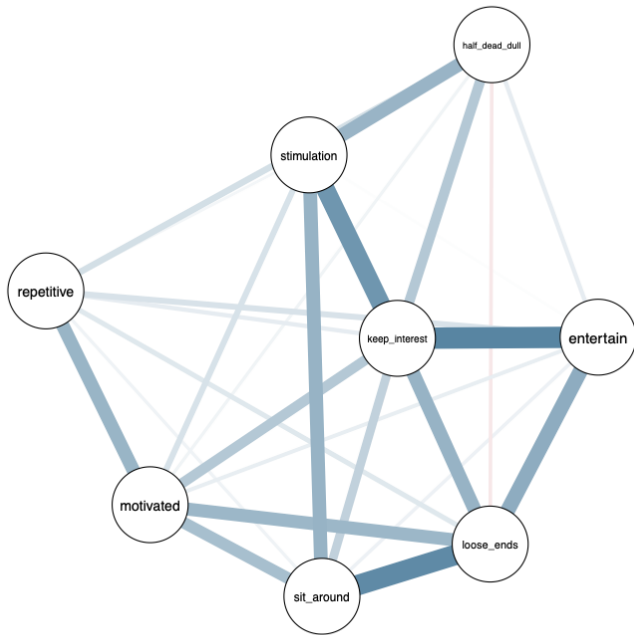


# Conditional independence?

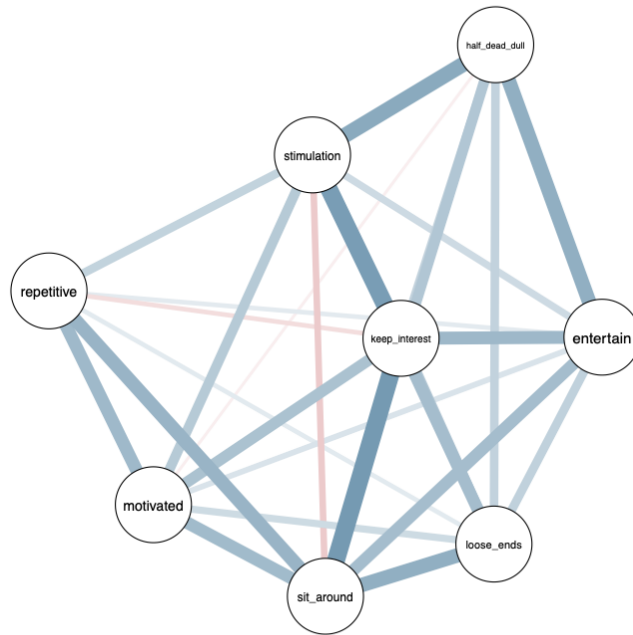
| EDGE SPECIFIC OVERVIEW |                        |          |                 |                 |       |              |
|------------------------|------------------------|----------|-----------------|-----------------|-------|--------------|
|                        | Relation               | Estimate | Posterior Incl. | Prob. Inclusion | BF    | Category     |
|                        | entertain-loose_ends   | 0.116    |                 | 1.000           | Inf   | included     |
|                        | repetitive-loose_ends  | 0.016    |                 | 0.353           | 0.546 | inconclusive |
|                        | stimulation-loose_ends | 0.079    |                 | 1.000           | Inf   | included     |
|                        | stimulation-entertain  | 0.001    |                 | 0.041           | 0.043 | excluded     |

$$\frac{\exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}{\sum_{\mathbf{x}} \exp \left( \sum_{i=1}^p \sum_{c=1}^{M_i} \mathcal{I}(x_i = c) \mu_{ic} + 2 \sum_{i>j} x_i x_j \sigma_{ij} \right)}$$

# French and English-Speaking Samples



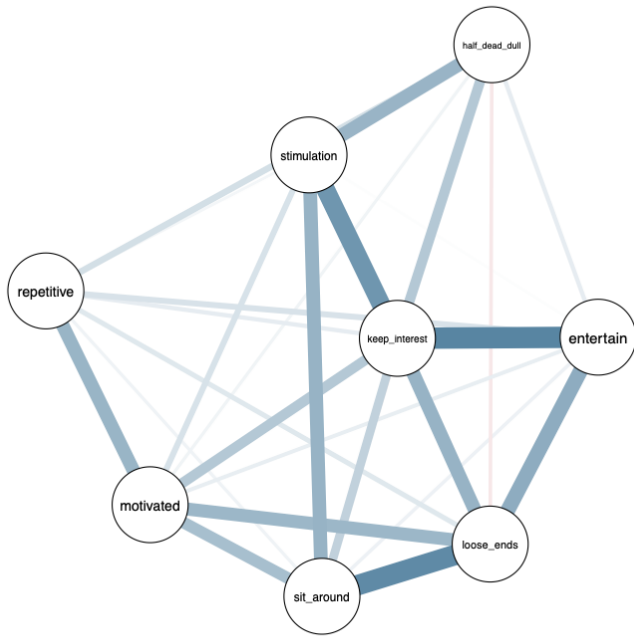
French (n = 496)



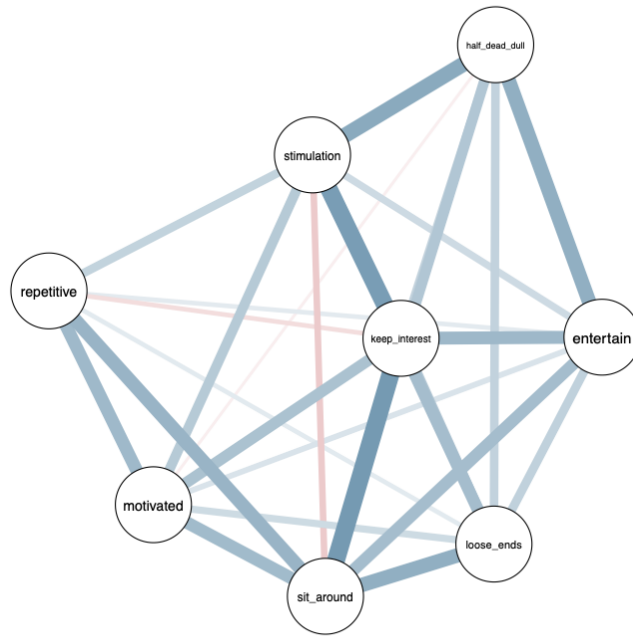
English (n = 490)

Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, 39(6), 390-399.

# French and English-Speaking Samples



French (n = 496)



English (n = 490)

How could we assess sample differences?

Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, 39(6), 390-399.

# French and English-Speaking Samples



French (n = 496)



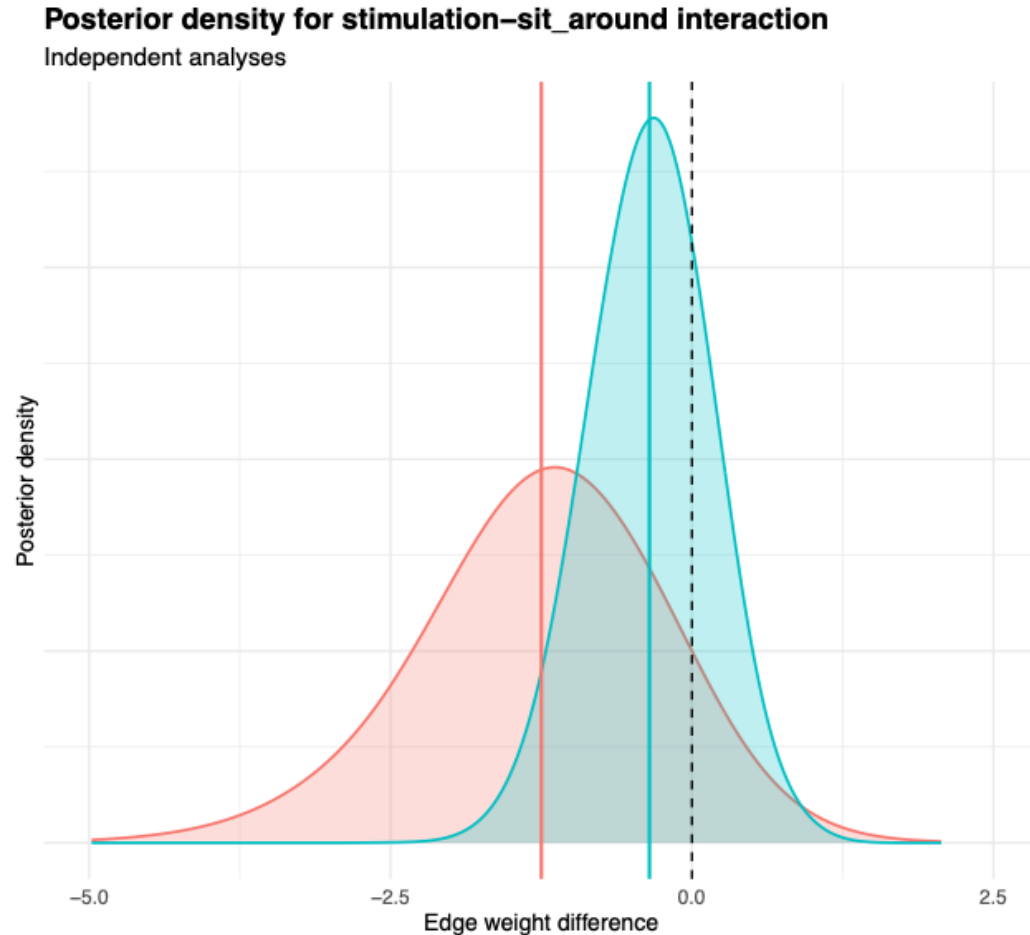
English (n = 490)

How could we assess sample differences?

Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, 39(6), 390-399.



# French and English-Speaking Samples



How could we assess sample differences?

Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, 39(6), 390-399.

# French and English-Speaking Samples



French (n = 496)

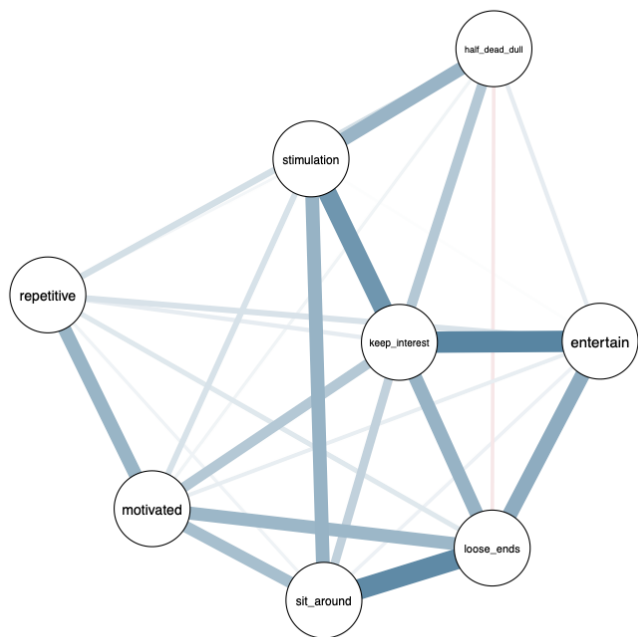


English (n = 490)

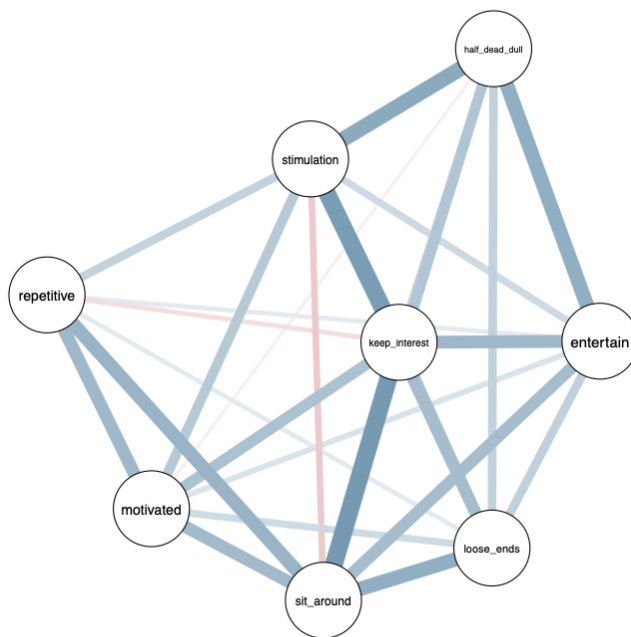
$$\sigma_{ij} = \begin{cases} \phi_{ij} - \frac{1}{2}\delta_{ij} & \text{for group 1} \\ \phi_{ij} + \frac{1}{2}\delta_{ij} & \text{for group 2} \end{cases}$$

Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, 39(6), 390-399.

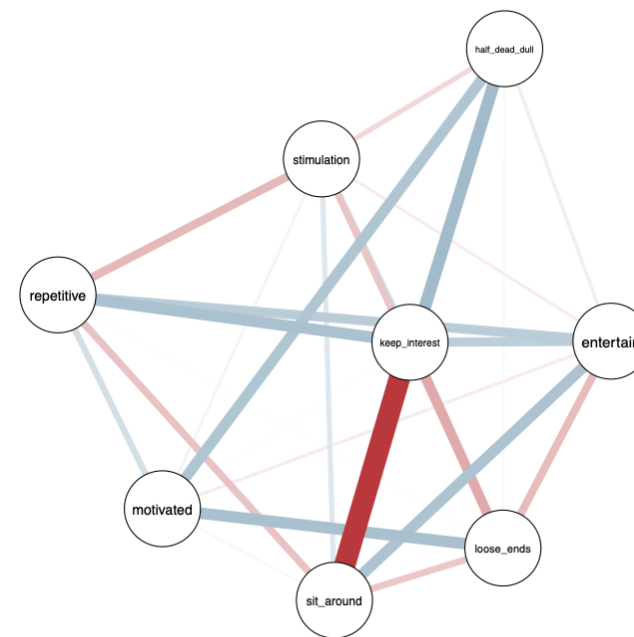
# Model the Difference in Edge Weight



French (n = 496)



English (n = 490)



$$\delta_{\text{diff}} = \sigma_{\text{diag}} - \sigma_{\text{no diag}}$$

# Test the Difference in Edge Weight

Rather than testing whether an edge weight should be included or excluded (i.e., conditional independence), we now test if an edge difference should be included or excluded (i.e., edge invariance).

# Test the Difference in Edge Weight

Rather than testing whether an edge weight should be included or excluded (i.e., conditional independence), we now test if an edge difference should be included or excluded (i.e., edge invariance).

$\mathcal{H}_0$ : Edge is equivalent in the two groups  $\iff \delta_{ij} = 0$

$\mathcal{H}_1$ : Edge is different in the two groups  $\iff \delta_{ij} \neq 0$

# Test the Difference in Edge Weight

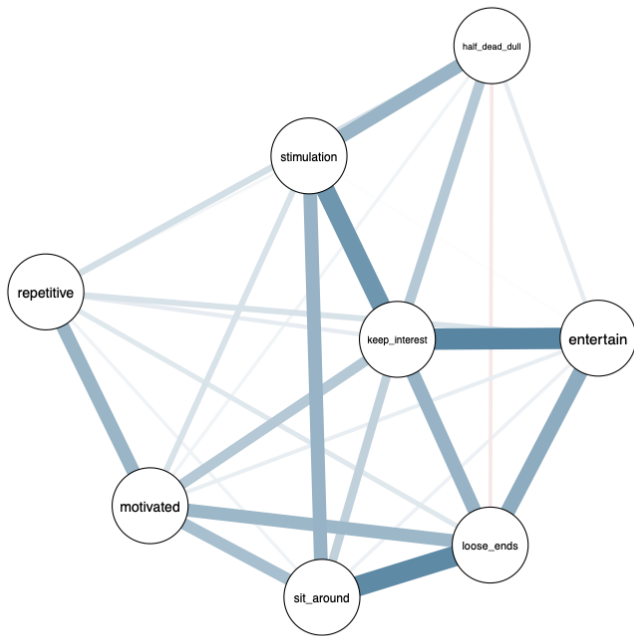
Rather than testing whether an edge weight should be included or excluded (i.e., conditional independence), we now test if an edge difference should be included or excluded (i.e., edge invariance).

$\mathcal{H}_0$ : Edge is equivalent in the two groups  $\iff \delta_{ij} = 0$

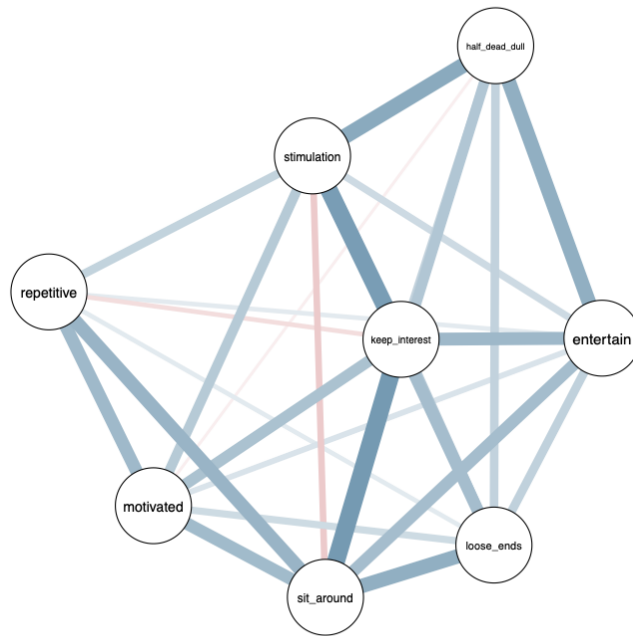
$\mathcal{H}_1$ : Edge is different in the two groups  $\iff \delta_{ij} \neq 0$

$$\text{BF}_{10} = \frac{p(\mathcal{H}_1 \mid \text{data})}{p(\mathcal{H}_0 \mid \text{data})} \bigg/ \frac{p(\mathcal{H}_1)}{p(\mathcal{H}_0)}$$

# Test the Difference in Edge Weight



French (n = 496)



English (n = 490)



# Classic vs Bayesian Network Comparison

## Classic Approaches

### Network Comparison Test (NCT)

- Permutation-based test
- Returns a  $p$ -value for rejecting equality

### Fused Graphical Lasso (FGL)

- Jointly estimates networks with penalization
- Encourages similarity across groups
- No inferential evidence for difference or invariance

## Limitations

### Asymmetry of evidence

Can reject equality, but not support it

### Multiplicity problems

Edge-wise testing inflates false positives or requires harsh corrections

### Visual fallacy

Missing edges are often interpreted as “no difference”



# Classic vs Bayesian Network Comparison

## Classic Approaches

### Network Comparison Test (NCT)

- Permutation-based test
- Returns a  $p$ -value for rejecting equality

### Fused Graphical Lasso (FGL)

- Jointly estimates networks with penalization
- Encourages similarity across groups
- No inferential evidence for difference or invariance

## Limitations

### Asymmetry of evidence

Can reject equality, but not support it

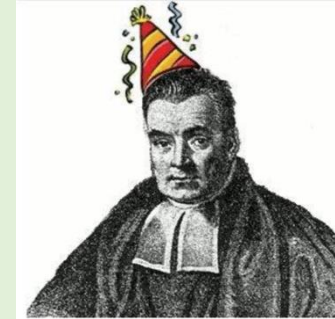
### Multiplicity problems

Edge-wise testing inflates false positives or requires harsh corrections

### Visual fallacy

Missing edges are often interpreted as “no difference”

## Bayesian Approaches



# Classic vs Bayesian Network Comparison

## Classic Approaches

### Network Comparison Test (NCT)

- Permutation-based test
- Returns a  $p$ -value for rejecting equality

### Fused Graphical Lasso (FGL)

- Jointly estimates networks with penalization
- Encourages similarity across groups
- No inferential evidence for difference or invariance

## Limitations

### Asymmetry of evidence

Can reject equality, but not support it

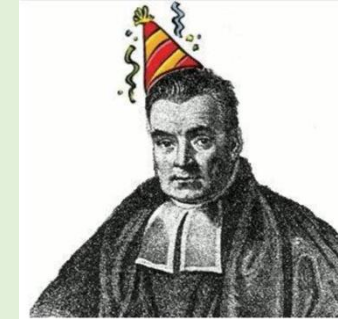
### Multiplicity problems

Edge-wise testing inflates false positives or requires harsh corrections

### Visual fallacy

Missing edges are often interpreted as “no difference”

## Bayesian Approaches



## Advantages

### Bayes factors

Quantify evidence *for differences* and *for invariance*

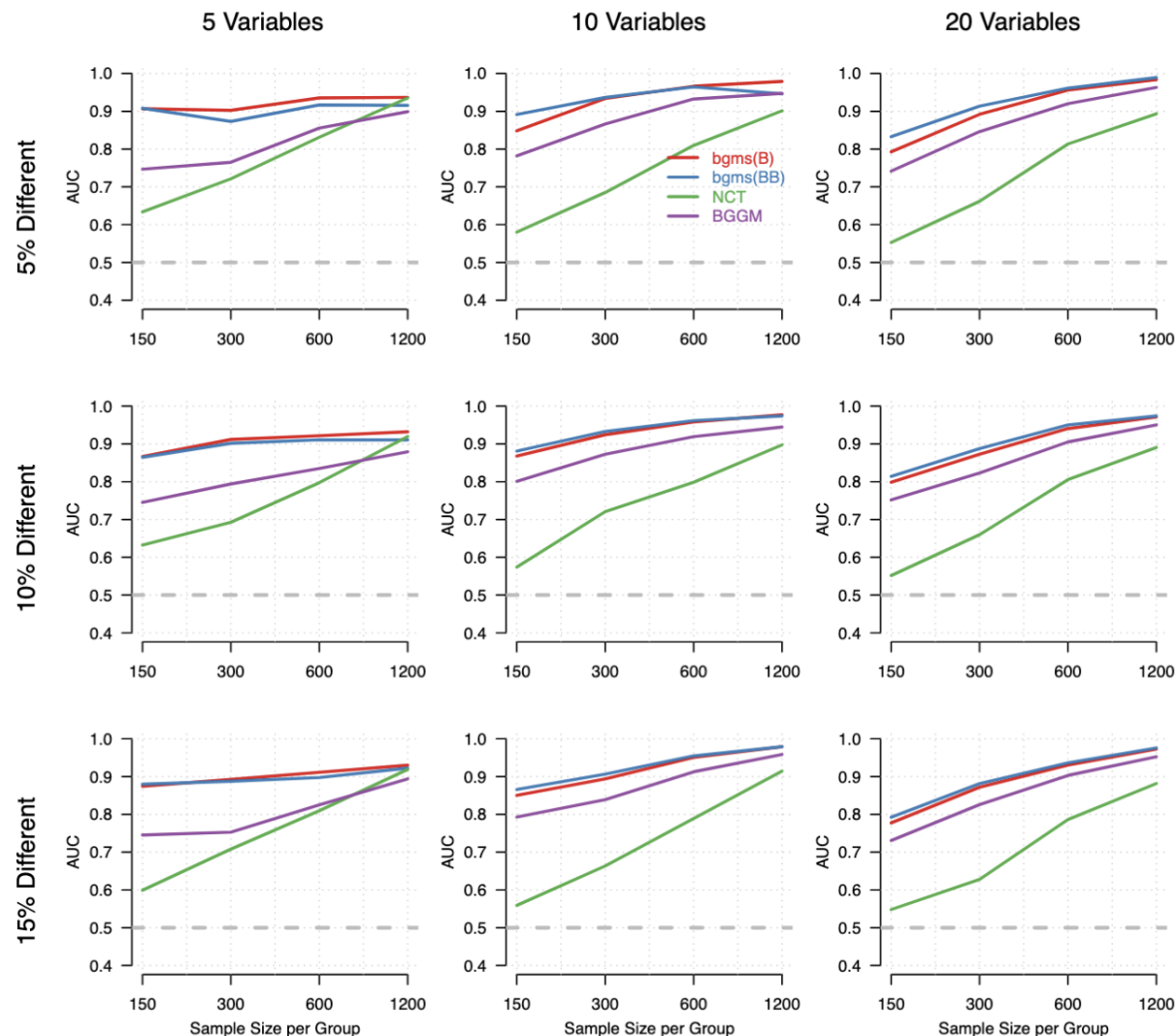
### Model averaging

Can automatically correct for multiplicity

### Posterior distributions and evidence plots

Make uncertainty explicit

# Simulation Evidence



## Area under the ROC curve

- Threshold-free summary of detection accuracy
- 0.5 = chance; higher = better

## Design:

- Variables: 5, 10, 20
- Edge differences: 5–15%
- Sample size: 150–1200 per group

## Results:

- Bayesian MRF methods perform best overall
- Strong gains in small samples and low signal
- NCT competitive with large samples & few variables

# Session setup



Part I: Theory



Part II: Practical



Part III: Project